

Foundations of machine learning

# Local asymptotic Normality

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Winter 2026

## The Normal means model as asymptotic approximation

- The Normal means model might seem quite special.
- But asymptotically, any sufficiently smooth parametric model is equivalent.
- Formally: The likelihood ratio process of  $n$  i.i.d. draws  $Y_i$  from the distribution

$$P_{\theta_0+h/\sqrt{n}}^n,$$

converges to the likelihood ratio process of one draw  $X$  from

$$N\left(h, I_{\theta_0}^{-1}\right)$$

- Here  $h$  is a local parameter for the model around  $\theta_0$ , and  $I_{\theta_0}$  is the Fisher information matrix.

- Suppose that  $P_\theta$  has a density  $f_\theta$  relative to some measure.
- Recall the following definitions:
  - Log-likelihood:  $\ell_\theta(Y) = \log f_\theta(Y)$
  - Score:  $\dot{\ell}_\theta(Y) = \partial_\theta \log f_\theta(Y)$
  - Hessian  $\ddot{\ell}_\theta(Y) = \partial_\theta^2 \log f_\theta(Y)$
  - Information matrix:  $I_\theta = \text{Var}_\theta(\dot{\ell}_\theta(Y)) = -E_\theta[\ddot{\ell}_\theta(Y)]$
- Likelihood ratio process:

$$\prod_i \frac{f_{\theta_0+h/\sqrt{n}}(Y_i)}{f_{\theta_0}(Y_i)},$$

where  $Y_1, \dots, Y_n$  are i.i.d.  $P_{\theta_0+h/\sqrt{n}}$  distributed.

## Practice problem (Taylor expansion)

- Using this notation, provide a second order Taylor expansion for the log-likelihood  $\ell_{\theta_0+h}(Y)$  with respect to  $h$ .
- Provide a corresponding Taylor expansion for the log-likelihood of  $n$  i.i.d. draws  $Y_i$  from the distribution  $P_{\theta_0+h/\sqrt{n}}$ .
- Assuming that the remainder is negligible, describe the limiting behavior (as  $n \rightarrow \infty$ ) of the log-likelihood ratio process

$$\log \prod_i \frac{f_{\theta_0+h/\sqrt{n}}(Y_i)}{f_{\theta_0}(Y_i)}.$$

## Solution

- Expansion for  $\ell_{\theta_0+h}(Y)$ :

$$\ell_{\theta_0+h}(Y) = \ell_{\theta_0}(Y) + h' \cdot \dot{\ell}_{\theta_0}(Y) + \frac{1}{2} \cdot h \cdot \ddot{\ell}_{\theta_0}(Y) \cdot h + \text{remainder}.$$

- Expansion for the log-likelihood ratio of  $n$  i.i.d. draws:

$$\log \prod_i \frac{f_{\theta_0+h'/\sqrt{n}}(Y_i)}{f_{\theta_0}(Y_i)} = \frac{1}{\sqrt{n}} h' \cdot \sum_i \dot{\ell}_{\theta_0}(Y_i) + \frac{1}{2n} h' \cdot \sum_i \ddot{\ell}_{\theta_0}(Y_i) \cdot h + \text{remainder}.$$

- Asymptotic behavior (by CLT, LLN):

$$\Delta_n := \frac{1}{\sqrt{n}} \sum_i \dot{\ell}_{\theta_0}(Y_i) \rightarrow^d N(0, I_{\theta_0}),$$
$$\frac{1}{2n} \cdot \sum_i \ddot{\ell}_{\theta_0}(Y_i) \rightarrow^p -\frac{1}{2} I_{\theta_0}.$$

- Suppose the remainder is negligible.
- Then the previous slide suggests

$$\log \prod_i \frac{f_{\theta_0+h/\sqrt{n}}(Y_i)}{f_{\theta_0}(Y_i)} \stackrel{A}{=} h' \cdot \Delta - \frac{1}{2} h' I_{\theta_0} h,$$

where

$$\Delta \sim N(0, I_{\theta_0}).$$

- Theorem 7.2 in van der Vaart (2000), chapter 7 states sufficient conditions for this to hold.
- We show next that this is the same likelihood ratio process as for the model

$$N\left(h, I_{\theta_0}^{-1}\right).$$

## Practice problem

- Suppose  $X \sim N(h, I_{\theta_0}^{-1})$
- Write out the log likelihood ratio

$$\log \frac{\varphi_{I_{\theta_0}^{-1}}(X - h)}{\varphi_{I_{\theta_0}^{-1}}(X)}.$$

## Solution

- The Normal density is given by

$$\varphi_{I_{\theta_0}^{-1}}(x) = \frac{1}{\sqrt{(2\pi)^k |\det(I_{\theta_0}^{-1})|}} \cdot \exp\left(-\frac{1}{2}x' \cdot I_{\theta_0} \cdot x\right)$$

- Taking ratios and logs yields

$$\log \frac{\varphi_{I_{\theta_0}^{-1}}(X - h)}{\varphi_{I_{\theta_0}^{-1}}(X)} = h' \cdot I_{\theta_0} \cdot x - \frac{1}{2}h' \cdot I_{\theta_0} \cdot h.$$

- This is exactly the same process we obtained before, with  $I_{\theta_0} \cdot X$  taking the role of  $\Delta$ .



## Why care

- Suppose that  $Y_i \sim^{iid} P_{\theta+h/\sqrt{n}}$ , and  $T_n(Y_1, \dots, Y_n)$  is an arbitrary statistic that satisfies

$$T_n \rightarrow^d L_{\theta,h}$$

for some limiting distribution  $L_{\theta,h}$  and all  $h$ .

- Then  $L_{\theta,h}$  is the distribution of some (possibly randomized) statistic  $T(X)$ !
- This is a (non-obvious) consequence of the convergence of the likelihood ratio process.
- cf. Theorem 7.10 in van der Vaart (2000).

## Maximum likelihood and shrinkage

- This result applies in particular to  $T =$  estimators of  $\theta$ .
- Suppose that  $\hat{\theta}^{ML}$  is the maximum likelihood estimator.
- Then  $\hat{\theta}^{ML} \rightarrow^d X$ , and any shrinkage estimator based on  $\hat{\theta}^{ML}$  converges in distribution to a corresponding shrinkage estimator in the limit experiment.

## References

*van der Vaart, A. W. (2000). Asymptotic statistics. Cambridge University Press, chapter 7.*

*Hansen, B. E. (2016). Efficient shrinkage in parametric models. Journal of Econometrics, 190(1):115–132.*